Longest Alternating Subsequence

Problem Statement

Given a number sequence, find the length of its Longest Alternating Subsequence (LAS). A subsequence is considered alternating if its elements are in alternating order.

A three element sequence (a1, a2, a3) will be an alternating sequence if its elements hold one of the following conditions:

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{a1 > a2 < a3 } or { a1 < a2 > a3}.
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Example 1:

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Input: {1,2,3,4}
Output: 2
Explanation: There are many LAS: {1,2}, {3,4}, {1,3}, {1,4}
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Example 2:

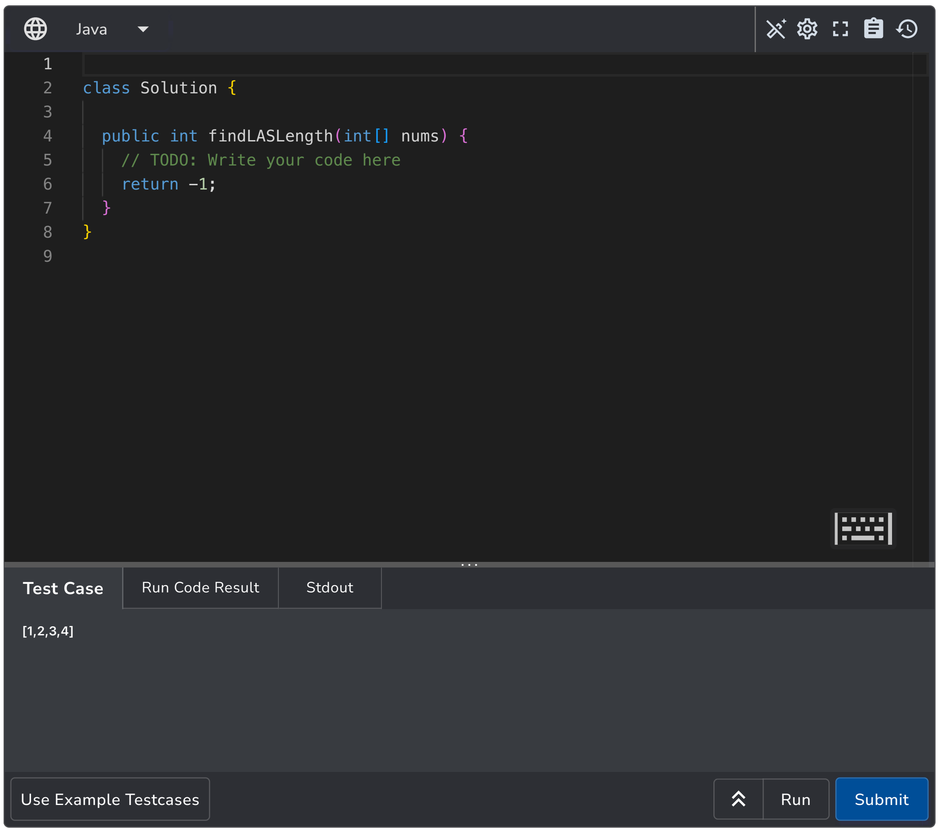
```
Input: {3,2,1,4}
Output: 3
Explanation: The LAS are {3,2,4} and {2,1,4}.
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Example 3:

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Input: {1,3,2,4}
Output: 4
Explanation: The LAS is {1,3,2,4}.
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